

- Filename: comptia-pentest-plus-pt0-002-5-1-1-basic-programming-concepts.md
- Show Name: Pentest+ (PT0-002)
- Topic Name: Tools and Code Analysis
- Episode Name: Basic Programming Concepts

Basic Programming Concepts

Objectives:

- List and define the basic programming concepts that make up most commonly used programming and scripting languages

- Logic Constructs

- Loops

- ```
var1=$(cat /etc/passwd | awk '{FS = ":"} {print $1}')
```

```
for i in $var1
```

```
do
```

```
echo "Username : " $i
```

```
done
```

- Conditional Statements

- Invoke Ruby with `irb`
    - `user1 = grep dlowrie /etc/passwd | cut -d ":" -f 1`

```
+ `if user1 == "dlowrie"`
+ `print "User found!"`
+ `else`
+ `print "User not found!"`
+ `end`
```

- Boolean Operator

- String Operator

- Arithmetic Operator

- Data Structures

- JSON (JavaScript Object Notation)

- ```
>>> import json
```

```
>>> x = '{ "name":"Dan", "age":41, "title":"Edutainer" }'
```

```
>>> y = json.loads(x)
```

- Dictionaries

- Key:Value pairs

- "brand": IProTV

- CSV

- Lists and Arrays

- Both store data
 - Both are changeable (mutable)
 - Both can be indexed and iterated through
 - Both can be sliced
 - Arrays are optimized for arithmetic operations

- Arrays are containers for elements of the same data type
- Arrays must be declared
- Lists are containers for elements with different data types
- Lists can be created ad-hoc

- Binary Search Trees

- Libraries

- Libraries are collections of pre-written code that users can use to optimize tasks.

- Classes

- Informs the attributes of an Object

```
class User:
    def __init__(self, Name, Account, SubLevel): #constructor
        self.name = Name
        self.account = Account
        self.sublevel = SubLevel

u1 = User("Ronnie", 1001, "Premium")
```

- Procedures and Functions

- Procedures

- List of commands executed in order
- Doesn't return a value or anything

- Functions

- List of commands executed in order
- Returns a value